Emilio Pisanty

ROYAL SOCIETY UNIVERSITY RESEARCH FELLOW AND PROLEPTIC LECTURER

Photonics & Nanotechnology Group · King's College London, UK

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Education

- **PhD in Controlled Quantum Dynamics**, Imperial College London, UK *2012-2016*
- **MRes in Controlled Quantum Dynamics** with Distinction, Imperial College London, UK *2011-2012*
- **Licenciatura en Física (BSc in Physics)** with Honours and 9.80/10 GPA, F. Ciencias, UNAM, Mexico *2006-2011*

Research Experience

King's College London *London*
ROYAL SOCIETY UNIVERSITY RESEARCH FELLOW AND PROLEPTIC LECTURER · Photonics & Nanotechnology group *2022 - present*

- Working on strong-field physics and attosecond science, and their intersections with structured light, catastrophe optics, and quantum electrodynamics

Max Born Institute for Nonlinear Optics and Short-Pulse Spectroscopy *Berlin*
WISSENSCHAFTLICHER MITARBEITER · group leader: Olga Smirnova *2020 - 2021*

- Settled the question on the existence optical skyrmions by providing explicit constructions

ICFO – The Institute of Photonic Sciences *Barcelona*
POSTDOCTORAL RESEARCHER · group leader: Maciej Lewenstein *2017 - 2020*

- Discovered and characterized a novel type of optical polarization vortex with new symmetries and topologies of light
- Designed the first nonlinear optical polarization tomography experiment, in collaboration with experimentalists
- Developed novel methods, and advanced existing ones, to analyze strong-field interactions for atoms and solids
- Directed the development, as an MSc supervisor, of a fully-quantum-optical framework for high-harmonic generation
- Discovered a natural definition of the cutoff for high-order harmonic generation

Max Born Institute for Nonlinear Optics and Short-Pulse Spectroscopy *Berlin*
WISSENSCHAFTLICHE MITARBEITER · group leader: Misha Ivanov *2016 - 2017*

- Designed a method to prevent the Lorentz-force suppression of high harmonics in the long-wavelength regime
- Implemented complex-trajectory methods on a rotating frame to understand the chirality of high-harmonic emission

Imperial College London *London*
POSTGRADUATE RESEARCHER · advisor: Misha Ivanov *2011 - 2016*

PhD Thesis: Electron dynamics in complex space and complex time
MRes Thesis: Under-the-barrier electron-ion interaction during tunnel ionization

- Characterized the role of the complex-valued quantum orbits in semiclassical treatments of strong-field ionization
- Used analytical tools to describe kinematical origins for Zero-Energy Structures in above-threshold ionization
- Clarified the conservation of spin angular momentum in high-order harmonic generation

Instituto de Ciencias Nucleares, UNAM *Mexico City*
ESTUDIANTE ASOCIADO · advisor: Eduardo Nahmad-Achar *2010 - 2011*

BSc Thesis: Generalized coherent states and the analytic structure of the annihilation operator

- Researched the quantum optics of field quadratures in a truncated photon basis, leading to a first-author undergraduate publication

Awards and fellowships

- King's Education Award (nomination), King's College London, 2025
- University Research Fellowship, Royal Society, 2022-2029
- Emerging Leaders 2021 award, *Journal of Optics*, 2021
- Cellex-ICFO-MPQ Postdoctoral Fellowship, Cellex Foundation, 2018-2020
- Rector's Scholarship Fund Award, Imperial College London, 2012 - 2015
- Sir Peter Knight prize for MRes performance, Centre for Doctoral Training in Controlled Quantum Dynamics, Imperial College London, 2012
- Mexican National Council for Science and Technology (CONACYT) foreign scholarship, 2011 - 2016
- Diploma for academic performance (second place of 283 by grade average), Facultad de Ciencias, Universidad Nacional Autónoma de México, 2011

RESEARCH FUNDING

- MSCA Postdoctoral Fellowship (as PI; fellow: Nicola Mayer), 2025-2026

- King's Undergraduate Research Fellowship (as PI; funding for one internship per award), King's College London, 2022; 2025 and 2026 (co-supervised with Anne Weber)
- Postgraduate Research International Fee Waiver for overseas PhD student, King's College London, 2022 - 2025
- Research Fellows Enhanced Research Expenses, Royal Society, 2022 - 2024 and 2023-2025
- Mexican Secretary of Education (SEP) postgraduate complementary scholarship, 2011 - 2012

Professional experience

Doctoral and postdoctoral alumni

- Anne Weber, PhD supervision, King's College London. Thesis: Caustics and catastrophes in strong-field processes: Picard-Lefschetz theory as a universal approach to saddle-point methods in attosecond science. 2022-2026
- Robert Jones, PDRA supervision, King's College London. 2023 - 2024
- Noslen Suárez, ICFO – The Institute of Photonic Sciences (contributions to supervision; supervisors: M Lewenstein & M Ciappina). Thesis: *Strong-field processes in atoms and polyatomic molecules*), 2017

Ongoing PhD and postdoctoral supervision

- Daijun Luo, Visiting researcher, King's College London. 2026
- Nicola Mayer, MSCA postdoctoral fellowship, King's College London. Project: TopROCS: Topologically protected chiral sensing. 2025 - present
- Elliott Boyt-Archibald, PhD student, King's College London. Project: Structurally-chiral media driven by strong classical and quantum dynamically-chiral light. 2025 - present
- Séraphine Oakley, PhD student, King's College London. Project: Spatial and temporal structures in attosecond light. 2023 - present
- Benjamin Miller, PhD student (as second supervisor), King's College London. Project: Tunable attosecond sources for transient x-ray coherent tomography. 2023 - present

MSc and undergraduate supervision

- Rachel Bohan, MSci student (co-supervised with M Khokhlova), King's College London. Project: Chiral response functions in high harmonic generation using structured continuum states in momentum representation. 2025 - 2026
- Kajol Mistry, MSci student (co-supervised with N Mayer and A Weber), King's College London. Project: The role of the initial orbital on high harmonic generation polarisation. 2025 - 2026
- Xiang Zhou, MSci student, King's College London. Project: Semi-classical model of quantum tunnelling. 2024 - 2025
- Pietro Polenghi and Charlotte Rixon, BSc 3rd-year project, King's College London. 2024 - 2025
- Yann Janssen, Jude Russell, Lyra So, Tom Casey, BSc 3rd-year project and King's Undergraduate Research Fellowship (YJ), King's College London. 2024 - 2025
- Shudi Mei, MSc student, King's College London. Project: Optimizing high-harmonic generation with customized polarizations and multi-driver lasers. 2023 - 2025
- Fedor Sokolov, MSci student (co-supervised with R Jones), King's College London. Project: Strong field effects in solids. 2023 - 2024
- Coco Kern, MSci student, King's College London. Project: Waves and beams in total internal reflection. 2023 - 2024
- Umar Akhtar, MSci student, King's College London. Project: Entanglement in Larmor clocks. 2023 - 2024
- Siu Tiung Wong, King's Undergraduate Research Fellow, King's College London. Project: Shaping electron trajectories by tailoring the polarization state of ultra-intense laser drivers. 2022
- Javier Rivera Deán, MSc student, ICFO. Thesis: Quantum-optical analysis of high-order harmonic generation. 2019
- Masudur Rahman, MSc student, ICFO. Thesis: Quantum simulations of attosecond physics), 2019

Teaching experience

- Co-lecturer, King's College London (50% of one module per academic year). Quantum Mechanics I (2023-2026), Introduction to Modern Physics (2022-2023). 2022 - present
- Designed and taught a course on Quantum Key Distribution for high-school students as part of the Barcelona International Youth Science Challenge, ICFO, 2019
- Developed and presented ICFO Theory Lectures series "Complex Analysis and Saddle-Point Methods for Strong-Field Physics and Beyond", ICFO, 2018
- Postgraduate Teaching Assistant, Imperial College London. Tutorials and Professional Skills seminars for 1st and 2nd year physics: Vector Calculus, Electromagnetism, Quantum Mechanics, Optics, Atomic Physics; Mathematica course for MRes students, 2012 - 2016
- Teaching Assistant, Universidad Nacional Autónoma de México. Analytic Geometry, Linear Algebra and Introduction to Electromagnetism; 1st and 2nd year physics and mathematics, 2010 - 2011

Open-Source Software

- Chimera: Chiral Measures Research Assistant, implementing measures of chirality via the 'chiral moments' formalism (GPL, CC-BY-SA)

- ComplexFocus: Non-paraxial vector beams in Mathematica, a flexible and robust implementation of analytical solutions for describing tightly-focused laser beams (GPL)
- LISSAFIRE: Lissajous-Figure Reconstruction for nonlinear polarization tomography of bichromatic fields (GPL, CC-BY-SA)
- RB-SFA: High Harmonic Generation in the Strong Field Approximation via Mathematica, a leading open-source implementation of the quantum-orbit and numerical-integration approaches to high-harmonic generation (GPL, CC-BY-SA)
- QuODD: Quantum-Orbits Dynamic Dashboard, for visualizing quantum orbits on complex time and complex space (MIT)
- ARMSupport, an open-source implementation of Analytical *R*-Matrix theory for atomic and molecular strong-field photoionization (GPL, CC-BY-SA)

Science communication

- Co-led the King's College London exhibit 'The Unseen Wonders of Light' at the Royal Society Summer Science Exhibition 2024 (with M Khokhlova, coordinator, and J Millen), including developing a brand-new demonstration of the 'optical barberpole' effect to present optical rotation by chiral molecules to a general-public audience.
- Active contributor at Physics Stack Exchange, with ∼2600 posts reaching an estimated audience of 5.3 million views, 2012-present. Samples: Chirped Pulse Amplification; Unit realizations in the New SI; Oscillating charge in the hydrogen atom
- Presented outreach talks at high-school level as part of the *Matí de la Reserca* 2019
- On-site coordinator with the Lightyear Foundation for the establishment of the Lab_13 Ghana practical-science teaching space, 2015
- Conducted practical-science education workshops in Ghana with the Lightyear Foundation, 2013
- Extensive science busking experience with Imperial College London and with the Lightyear Foundation, including the Green Man festival, the Big Bang Fair, and the Cheltenham Literature Festival; 2012-2016
- Presented outreach talks "Boat wakes", 2012, and "Chirp", 2014, Imperial College London

Professional service

- Programme Committee, International Conference on Atomic Physics 2024
- 43 journal peer reviews, including *Phys. Rev. A*, *Nat. Commun.*, *New J. Phys.*, *J. Phys. B*, *Nanophotonics*, *Phys. Rev. Lett.*, *Phys. Rev. X*, *Nat. Phys.* and *PRX Quantum* (Web of Science reviewer profile link), 2013 - present
- PhD thesis examiner for two PhD thesis (Universidad Autónoma de Madrid, 2025; University of Jena, 2023)
- IOP Trusted Reviewer since 2024
- Student editor for the Centres for Doctoral Training Newsletter, Imperial College London, 2012
- Organizing committee, Summer School on Quantum Information, Computing and Control (QuICC) 2012

Additional experience

- Experience with web-based 3D figures and with 3D printing for visualization, communication and publication of complex geometrical concepts
- Experience with Wolfram Mathematica in both research, package development, and teaching roles
- Developed LaTeX article and poster templates; experience with git and mercurial for software and project management
- Black belt in Kung-fu Wu-shu, 1st and 2nd degree. Asociación Calmecac, Mexico City. 2006, 2008
- Bilingual in Spanish and English, intermediate-level German, beginner Russian and French

Publications

Selected publications

- A Weber, J Feldbrugge, E Pisanty. A universal approach to saddle-point methods in attosecond science. *Phys. Rev. A*, in press (2026), arXiv:2510.12545

  Theoretical work by PhD student Anne Weber, presenting a rigorous foundation for ubiquitous saddle-point methods in attosecond science, universally applicable to arbitrary driving waveforms as well as to oscillatory integrals of general nature.

- M Lewenstein, MF Ciappina, E Pisanty, J Rivera-Dean, T Lamprou, P Tzallas. Generation of optical Schrödinger cat states in intense laser–matter interactions. *Nat. Phys.* 17, 1104 (2021), arXiv:2008.10221

  Experimental and theoretical collaboration showing the measurement of the Wigner function of an intense laser pulse, together with other quantum properties of the light pulse, after it has driven high-harmonic generation in a gas jet.

- R Gutiérrez-Cuevas, E Pisanty. Optical polarization skyrmionic fields in free space. *J. Opt.* 2, 024004 (2021), arXiv:2101.09254

  Work on structured light inspired by strong-field and attosecond physics, showing skyrmion structures discovered in optical polarization vortices during a search for exotic polychromatic topologies of light. Published in the Emerging Leaders 2021 collection of *J. Opt.*

- E Pisanty, MF Ciappina and M Lewenstein. The imaginary part of the high-harmonic cutoff. *J. Phys: Photon.* 2, 034013 (2020), arXiv:2003.00277,

  Presents, for the first time in 35 years, a rigorous and universal definition for the harmonic cutoff for high-harmonic generation, showing that it is the 'bifurcation set' of a caustic, and constructing a natural imaginary part for the cutoff which controls the interference between quantum pathways near the cutoff.

- L Rego, KM Dorney, NJ Brooks, Q Nguyen, C-T Liao, J San Román, DE Couch, A Liu, E Pisanty, M Lewenstein, L Plaja, HC Kapteyn, MM Murnane and C Hernández-García. Generation of extreme-ultraviolet beams with time-varying orbital angular momentum. *Science* 364, aaw9486 (2019), arXiv:1901.10942

  Highest-cited paper, and covered in a variety of news outlets including National Geographic, New Scientist and El País. Theoretical and experimental collaboration that used the time-domain perspectives of ultrafast science to discover the 'self-torque', a novel property of light.

- E Pisanty, GJ Machado, V Vicuña-Hernández, A Picón, A Celi, JP Torres and M Lewenstein. Knotting fractional-order knots with the polarization state of light. *Nat. Photon.* 13, 569 (2019), arXiv:1808.05193

  Presents novel symmetries, topologies, and conserved quantities in light that are only seen when one studies light using polychromatic optics from the time-domain perspective.

- E Pisanty, L Rego, J San Román, A Picón, KM Dorney, HC Kapteyn, MM Murnane, L Plaja, M Lewenstein and C Hernández-García. Conservation of torus-knot angular momentum in high-order harmonic generation. *Phys. Rev. Lett.* 122, 203201 (2019), arXiv:1810.06503

  Shows the rich structures of light that arise in high-harmonic generation driven by complex polychromatic optical polarization vortices.

- E Pisanty and M Ivanov. Slalom in complex time: emergence of low-energy structures in tunnel ionization via complex time contours. *Phys. Rev. A* 94, 043408 (2016), arXiv:1507.00011

  Core theory-development PhD paper, exploring the interplay between the atomic Coulomb potential and the complex-time trajectory dynamics that arise in semiclassical treatments of strong-field physics, and showing that the position itself must also be considered as a complex quantity.

Full publication list

51 publications (44 peer-reviewed) with h-index 27 (Google Scholar) / 21 (Web of Science) / 21 (Scopus). Total citation count: 2891 (Google Scholar) / 1858 (Web of Science) / 1943 (Scopus), cited in 1313 (WOS) / 1373 (Scopus) publications.

- X Zou, L Jurkovičová, A Weber, C Zhao, M Albrecht, O Finke, A Vendl, A Grenfell, W Szuba, J Nejdl, E Constant, M Khokhlova, E Pisanty, O Hort, A Zaïr. 2D quantum-path interference in high-harmonic generation driven by highly-bichromatic fields. arXiv:2604.12838 (2026)
- E Pisanty, N Mayer, A Ordóñez, A Löhr, M Khokhlova. Chiral moments make chiral measures. arXiv:2603.26793 (2026)
- A Weber, J Feldbrugge, E Pisanty. A universal approach to saddle-point methods in attosecond science. *Phys. Rev. A*, in press (2026), arXiv:2510.12545
- J Rivera-Dean, Th Lamprou, E Pisanty, MF Ciappina, P Tzallas, M Lewenstein, P Stammer. Quantum state engineering of light using intensity measurements and postselection *Phys. Rev. A* 112, 013110 (2025), arXiv:2409.02016
- A Weber, M Khokhlova, E Pisanty. Quantum tunnelling without a barrier. *Phys. Rev. A* 111, 043103 (2025), arXiv:2311.14826
- J Rivera-Dean, HB Crispin, P Stammer, Th Lamprou, E Pisanty, M Krüger, P Tzallas, M Lewenstein, MF Ciappina. Squeezed states of light after high-harmonic generation in excited atomic systems. *Phys. Rev. A* 110, 063118 (2024), arXiv:2408.07577
- JFP Mosquera, G Cistaro, M Malakhov, E Pisanty, A Dauphin, L Plaja, A Chacón, M Lewenstein, A Picón. Topological phase transitions via attosecond x-ray absorption spectroscopy. *Rep. Progr. Phys.* 87, 117907 (2024), arXiv:2407.03737
- J Rivera-Dean, Th Lamprou, E Pisanty, MF Ciappina, P Tzallas, M Lewenstein, P Stammer Quantum state engineering of light using intensity measurements and post-selection. *Phys. Rev. A* 112, 013110 (2025), arXiv:2409.02016
- N Mayer, D Ayuso, P Decleva, M Khokhlova, E Pisanty, M Ivanov, O Smirnova. Chiral topological light for detection of robust enantiosensitive observables. *Nat. Photon.* 18, 1155 (2024), arXiv:2303.10932
- J Rivera-Dean, P Stammer, AS Maxwell, T Lamprou, AF Ordóñez, E Pisanty, P Tzallas, M Lewenstein, MF Ciappina. Quantum Optical Analysis of High-Order Harmonic Generation in Semiconductors. In High-Order Harmonic Generation in Solids (M. Ciappina and P. Tzallas, Eds.), pp. 139-183 (World Scientific, 2024)
- J Rivera-Dean, P Stammer, AS Maxwell, T Lamprou, E Pisanty, P Tzallas, M Lewenstein, MF Ciappina. Quantum optical analysis of high-order harmonic generation in H_2^+ molecules. *Phys. Rev. A* 109, 033706 (2024), arXiv:2307.12381
- M Lewenstein, N Baldelli, U Bhattacharya, J Biegert, MF Ciappina, U Elu, T Grass, PT Grochowski, A Johnson, Th Lamprou, AS Maxwell, A Ordóñez, E Pisanty, J Rivera-Dean, P Stammer, I Tyulnev, P Tzallas. Attosecond physics and quantum information science. *Proc. 8th Int. Conf. Attosecond Sci. Tech. ATTO 2023. Springer Proc. Phys.* 300, 27 (2043), arXiv:2208.14769
- J Rivera-Dean, P Stammer, AS Maxwell, T Lamprou, AF Ordóñez, E Pisanty, P Tzallas, M Lewenstein, MF Ciappina. Non-classical states of light after high-harmonic generation in semiconductors: A Bloch-based perspective. *Phys. Rev. B* 109, 035203 (2024), arXiv:2309.14435
- A Weber, M Khokhlova, E Pisanty. Quantum tunnelling without a barrier. arXiv:2311.14826(2023)
- M Khokhlova, E Pisanty, A Zaïr. Shining the shortest flashes of light on the secret life of electrons. *Adv Phot.* 6, 060501 (2023) [Invited News & Commentary]

- U Bhattacharya, T Lamprou, AS Maxwell, AF Ordóñez, E Pisanty, J Rivera-Dean, P Stammer, MF Ciappina, M Lewenstein, P Tzallas. Strong–laser–field physics, non–classical light states and quantum information science. *Rep. Prog. Phys.* 86, 094401 (2023), arXiv:2302.04692
- M Luttmann, M Vimal, M Guer, J-F Hergott, AZ Khoury, C Hernández-García, E Pisanty, T Ruchon. Conservation of a half-integer angular mmentum in nonlinear optics with a polarization Möbius strip. *Sci. Adv.* 9, adf3486 (2023), arXiv:2209.00454
- J Rivera-Dean, P Stammer, AS Maxwell, T Lamprou, AF Ordóñez, E Pisanty, P Tzallas, M Lewenstein, MF Ciappina. Entanglement and non-classical states of light in a strong-laser driven solid-state system. arXiv:2211.00033(2022)
- M Khokhlova, E Pisanty, S Patchkovskii, O Smirnova, M Ivanov. Enantiosensitive steering of free-induction decay. *Sci. Adv.* 8, eabq1962 (2022), arXiv:2109.15302
- P Stammer, J Rivera-Dean, T Lamprou, E Pisanty, MF Ciappina, P Tzallas, M Lewenstein. High photon number entangled states and coherent state superposition from the extreme-ultraviolet to the far infrared. *Phys. Rev. Lett.* 128, 123603 (2022), arXiv:2107.12887
- J Rivera-Dean, T Lamprou, E Pisanty, P Stammer, AF Ordóñez, AS Maxwell, MF Ciappina, M Lewenstein, P Tzallas. Quantum optics of strongly laser–driven atoms and generation of high photon number optical cat states. *Phys. Rev. A* 105, 033714 (2022), arXiv:2110.01032
- J Rivera-Dean, P Stammer, E Pisanty, T Lamprou, P Tzallas, M Lewenstein, MF Ciappina. New schemes for creating large optical Schrödinger cat states using strong laser fields. *J. Comput. Electron.* 128, 2111 (2021), arXiv:2107.12811
- M Lewenstein, MF Ciappina, E Pisanty, J Rivera-Dean, T Lamprou, P Tzallas. Generation of optical Schrödinger cat states in intense laser–matter interactions. *Nat. Phys.* 17, 1104 (2021), arXiv:2008.10221
- Emilio Pisanty. Knotted topologies in the polarization state of bichromatic light. Proc. SPIE 11818, Laser Beam Shaping XXI 11818809 (2021) [conference proceedings].
- GSJ Armstrong, MA Khokhlova, M Labeye, AS Maxwell, E Pisanty, M Ruberti. Dialogue on analytical and *ab initio* methods in attoscience. *Eur. Phys. J. D* 75, 209 (2021), arXiv:2102.07453
- Y Kang, E Pisanty, M Ciappina, M Lewenstein, C Figueira de Morisson Faria, AS Maxwell. Conservation laws for electron vortices in strong-field ionisation. *Eur. Phys. J. D* 75, 199 (2021), arXiv:2102.07453
- Th Lamprou, R Lopez-Martens, S Haessler, I Liontos, S Kahaly, J Rivera-Dean, P Stammer, E Pisanty, MF Ciappina, M Lewenstein, P Tzallas. Quantum-optical spectrometry in relativistic laser–plasma interactions using the high-harmonic generation process: a proposal. *Photonics* 8, 192 (2021)
- EG Neyra, P Vaveliuk, E Pisanty, AS Maxwell, M Lewenstein, MF Ciappina. Principal frequency of an ultrashort laser pulse. *Phys. Rev. A* 103, 053124 (2021), arXiv:2101.10526
- R Gutiérrez-Cuevas, E Pisanty. Optical polarization skyrmionic fields in free space. *J. Opt.* 2, 024004 (2021), arXiv:2101.09254
- AP Woźniak, M Lesiuk, DK Efimov, M Mandrysz, JS Prauzner-Bechcicki, M Ciappina, E Pisanty, J Zakrzewski, M Lewenstein, R Moszyński. A systematic construction of Gaussian basis sets for the description of laser field ionization and high-harmonic generation. *J. Chem. Phys.* 154, 094111 (2021), arXiv:2007.10375
- AS Maxwell, GSJ Armstrong, MF Ciappina, E Pisanty, Y Kang, AC Brown, M Lewenstein, C Figueira de Morisson Faria. Manipulating twisted electrons in strong-field ionization. *Faraday Discuss.* 228, 394 (2021), arXiv:2010.08355
- A Chacón, D Kim, W Zhu, SP Kelly, A Dauphin, E Pisanty, AS Maxwell, A Picón, MF Ciappina, DE Kim, C Ticknor, A Saxena and M Lewenstein. Circular dichroism in high-order harmonic generation: Heralding topological phases and transitions in Chern insulators. *Phys. Rev. B* 102, 134115 (2020), arXiv:1807.01616
- E Pisanty, MF Ciappina and M Lewenstein. The imaginary part of the high-harmonic cutoff. *J. Phys: Photon.* 2, 034013 (2020), arXiv:2003.00277
- S Mitra, S Biswas, J Schötz, E Pisanty, B Förg, GA Kavuri, C Burger, W Okell, M Högner, I Pupeza, V Pervak, M Lewenstein, P Wnuk and MF Kling. Suppression of individual peaks in two-colour high harmonic generation *J. Phys. B: At. Mol. Opt. Phys.* 53, 134004 (2020), arXiv:2007.15450
- J Schoetz, Z Wang, E Pisanty, M Lewenstein, MF Kling and MF Ciappina. Perspective on Petahertz Electronics and Attosecond Nanoscopy. *ACS Photonics* 6, 3057 (2019), arXiv:1912.08574
- K Amini, J Biegert, F Calegari, A Chacón, MF Ciappina, A Dauphin, DK Efimov, C Figueira de Morisson Faria, K Giergiel, P Gniewek, AS Landsman, M Lesiuk, M Mandrysz, AS Maxwell, R Moszyński, L Ortmann, JA Pérez-Hernández, A Picón, E Pisanty, J Prauzner-Bechcicki, K Sacha, N Suárez, A Zaïr, J Zakrzewski and M Lewenstein. Symphony on Strong Field Approximation. *Rep. Prog. Phys.* 82, 116001 (2019), arXiv:1812.11447
- L Rego, KM Dorney, NJ Brooks, Q Nguyen, C-T Liao, J San Román, DE Couch, A Liu, E Pisanty, M Lewenstein, L Plaja, HC Kapteyn, MM Murnane and C Hernández-García. Generation of extreme-ultraviolet beams with time-varying orbital angular momentum. *Science* 364, aaw9486 (2019), arXiv:1901.10942
- E Pisanty, GJ Machado, V Vicuña-Hernández, A Picón, A Celi, JP Torres and M Lewenstein. Knotting fractional-order knots with the polarization state of light. *Nat. Photon.* 13, 569 (2019), arXiv:1808.05193
- E Pisanty, L Rego, J San Román, A Picón, KM Dorney, HC Kapteyn, MM Murnane, L Plaja, M Lewenstein and C Hernández-García. Conservation of torus-knot angular momentum in high-order harmonic generation. *Phys. Rev. Lett.* 122, 203201 (2019), arXiv:1810.06503
- VE Nefedova, MF Ciappina, O Finke, M Albrecht, J Vábek, M Kozlová, N Suárez, E Pisanty, M Lewenstein and J Nejdl. Determination of the spectral variation origin in high-order harmonic generation in noble gases. *Phys. Rev. A* 98, 033414

(2018), arXiv:1806.03974
- E Pisanty, D Hickstein, BR Galloway, CG Durfee, HC Kapteyn, MM Murnane and M Ivanov. High harmonic interferometry of the Lorentz force in strong mid-infrared laser fields. *New J. Phys.* 20, 053036 (2018), arXiv:1606.01931
- N Suárez, A Chacón, E Pisanty, L Ortmann, AS Landsman, A Picón, J Biegert, M Lewenstein and MF Ciappina. Above-threshold ionization in multicenter molecules: the role of the initial state. *Phys. Rev. A* 97, 033415 (2018), arXiv:1709.04366
- Á Jiménez-Galán, N Zhavoronkov, D Ayuso, F Morales, S Patchkovskii, M Schloz, E Pisanty, O Smirnova and M Ivanov. Control of attosecond light polarization in two-color bi-circular fields. *Phys. Rev. A* 97, 023409 (2017), arXiv:1805.02250
- E Pisanty and Á Jiménez-Galán. Strong-field approximation in a rotating frame: high-order harmonic emission from *p* states in bicircular fields. *Phys. Rev. A* 96, 063401 (2017), arXiv:1709.00397
- BR Galloway, D Popmintchev, E Pisanty, DD Hickstein, MM Murnane, HC Kapteyn and T Popmintchev. Lorentz drift compensation in high harmonic generation in the soft and hard X-ray regions of the spectrum. *Opt. Express* 24, 21818 (2016)
- E Pisanty and M Ivanov. Kinematic origin for near-zero energy structures in mid-IR strong field ionization. *J. Phys. B: At. Mol. Opt. Phys.* 49, 105601 (2016)
- E Pisanty and M Ivanov. Slalom in complex time: emergence of low-energy structures in tunnel ionization via complex time contours. *Phys. Rev. A* 94, 043408 (2016), arXiv:1507.00011
- E Pisanty, S Sukiasyan and M Ivanov. Spin conservation in high-order-harmonic generation using bicircular fields. *Phys. Rev. A* 90, 043829 (2014), arXiv:1404.6242
- M Ivanov and E Pisanty. High-harmonic generation: Taking control of polarization. *Nat. Photon.* 8, 501 (2014) [News & Views]
- E Pisanty and M Ivanov. Momentum transfers in correlation-assisted tunnelling. *Phys. Rev. A* 89, 043416 (2014), arXiv:1307.4765
- E Pisanty and E Nahmad-Achar. On the spectrum of field quadratures for a finite number of photons. *J. Phys. A: Math. Theor.* 45, 395303 (2012), arXiv:1109.5724

Presentations

Invited conference presentations

- Knotted topologies in the polarization state of bichromatic light. Conference on Research and Innovations in Science and Technology of Materials (CRISTMAS), Paris, 2025
- Knotted topologies in the polarization state of bichromatic light. 17th International Conference "Correlation Optics", Chernivtsi, Ukraine (hybrid format), 2025
- Quantum-orbit dynamics in strongly-polychromatic laser fields. Strong-field Physics Encounters Quantum ElectroDynamics (SPEQED), MPI-PKS, Dresden, 2025
- Knotted topologies in the polarization state of bichromatic light. 15th International Conference on Metamaterials, Photonic Crystals and Plasmonics (META 2025), Málaga, 2025
- Quantum-orbit dynamics in strongly-polychromatic laser fields. 10th International Conference on Attosecond Science and Technology (ATTO X), Lund, Sweden, 2025
- Optical Tunnelling without a Barrier? Gordon Research Conference on Multiphoton Processes, Providence, Rhode Island, 2024
- Novel symmetries and topologies in polychromatic singular optics. Workshop on Structured Light and Multimode Systems, Rovaniemi, Finland, 2024
- Optical Tunnelling without a Barrier? 53rd Winter Colloquium on the Physics of Quantum Electronics (PQE24), Snowbird, UTAH, 2024
- Knotted topologies in the polarization state of bichromatic light. Control of Ultrafast Processes Using Structured Light (CUPUSL), MPI-PKS, Dresden, 2023
- Knotted topologies in the polarization state of bichromatic light. Invited colloquium, Department of Physics, University of Glasgow, 2023
- Novel symmetries, topologies and conservation laws revealed by high-harmonic generation in polychromatic optical vortices. 52nd Winter Colloquium on the Physics of Quantum Electronics (PQE23), Snowbird, UTAH, 2023
- The imaginary part of the high-order harmonic cutoff. International Workshop on Atomic Physics, MPI-PKS, Dresden, 2022
- The imaginary part of the high-order harmonic cutoff. 30th Annual International Laser Physics Workshop, online, 2022
- Knotted topologies in the polarization state of bichromatic light. Laser Beam Shaping XXI, hybrid (San Diego, online), 2021
- *Ab-initio* vs analytical methods. Invited 'battle' round-table discussion, Quantum Battles in Attoscience, online, 2020
- Creating and multiplying knotted topologies in the polarization state of light. International Workshop on Atomic Physics, MPI-PKS, 2019
- Slalom in complex time: semiclassical trajectories in strong-field ionization and their analytical continuations. BIRS Workshop on Mathematical and Numerical Methods for Time-Dependent Quantum Mechanics, Oaxaca, 2017.

Oral presentations

- Chiral Moments make Chiral measures. Contributed talk, Conference on Extreme Light and Chiral Molecular Systems (ELCH), Kassel, Germany, 2025
- Optical Tunnelling without a Barrier? Contributed talk, Photon 2024, Swansea, 2024
- The imaginary part of the high-order harmonic cutoff. Poster Highlight, 8th International Conference on Attosecond Science and Technology (ATTO VIII), Orlando, Florida, 2022
- Three-dimensional polychromatic knots and skyrmionic textures via tightly-focused beams. 6th International Conference on Optical Angular Momentum, Tampere, Finland, 2022
- The imaginary part of the high-order harmonic cutoff. International Conference on Photonic, Electronic and Atomic Collisions (VICPEAC), online, 2021
- Conservation of Torus-Knot Angular Momentum in High-Harmonic Generation Driven by Fields with Spin-Orbit Mixing. DAMOP 2020, online, 2020
- Conservation of Torus-Knot Angular Momentum in High-Harmonic Generation Driven by Fields with Spin-Orbit Mixing. USTS 2019 Meeting, Madrid, 2019
- Knotted topologies in the polarization state of bichromatic light. 5th International Conference on Optical Angular Momentum (ICOAM), Ottawa, 2019
- Knotting fractional-order knots with light's polarization. ICFODay Research Talk, 2017; Contributed talk, QUTIF Young Researcher meeting, 2018; Invited Seminar, MPQ, CEA-Saclay, UCL, King's College London, Imperial College London, Lund University, 2018 - 2020.
- Haces de luz con nudos de polarización. Invited Seminar, University of Salamanca, 2017; Invited Seminar, ICN-UNAM, 2018
- Slalom in complex time: semiclassical trajectories in strong-field ionization and their analytical continuations. Invited Seminar, ICN-UNAM, 2017
- High-harmonic interferometry of the Lorentz force in strong mid-IR laser fields. QUTIF Young Researcher Meeting, Göttingen, 2016
- Complex trajectories for quantum orbits. Invited seminar, Rostock University, 2016
- Electron dynamics in complex time and complex space. ICFO invited seminar, 2016
- Probing non-dipole effects in HHG using noncollinear beams. XLIC Meeting, Belgium, 2016
- Complex time contours in tunnel ionization and low-energy structures. APS March Meeting, San Antonio, 2015
- Spin conservation in bicircular HHG. 1st XLIC WG1 meeting, University College London„ 2014
- Spin conservation in bicircular HHG: a photon exchange model. 5th Taller de Estructura de la Materia, ICF-UNAM, 2014
- The role of correlations in tunnel ionization. Invited seminar, Azpuru-Guzik group, Harvard University, 2013
- Momentum transfers in correlation-assisted tunnel ionization. 3rd AQuA Student Congress, Imperial College, 2013
- Interactions during the tunnel effect. II Symposium of CONACYT Scholars and Ex-scholars in Europe, Strasbourg, 2012

Poster presentations

- A Nonlinear Optical Chirality for High-Harmonic Generation. Conference on Extreme Light and Chiral Molecular Systems (ELCH), Kassel, 2025; 10th International Conference on Attosecond Science and Technology (ATTO X), Lund, Sweden, 2025
- Optical Tunnelling without a Barrier? 9th International Conference on Attosecond Science and Technology (ATTO IX), Jeju, Korea, 2023
- The imaginary part of the high-order harmonic cutoff. 8th International Conference on Attosecond Science and Technology (ATTO VIII), Orlando, Florida, 2022
- The imaginary part of the high-order harmonic cutoff. DAMOP 2020, online, 2020
- Conservation of torus-knot angular momentum in high-harmonic generation driven by fields with spin-orbit mixing. ATTO2019, Szeged, 2019. 7th International Conference on Attosecond Science and Technology (ATTO VII), Szeged, Hungary, 2019
- Creating and multiplying knots in the polarization state of light. ICAP, Barcelona, 2018
- Conservation of torus-knot angular momentum in high-harmonic generation. GRC Multiphoton Processes, Providence, 2018
- Anatomy of high-order harmonic emission from *p* states in bicircular fields. ICOMP 14, Budapest, 2017
- Recovering high-harmonic emission from Lorentz-force effects using noncollinear counter-rotating beams. International Workshop on Atomic Physics, MPI-PKS, 2016
- Slalom in complex time: dealing with the imaginary position of a quantum orbit. ATTO2015, Saint-Sauveur, 2015; 3rd XLIC General Meeting, Debrecen, 2015; International Workshop on Atomic Physics, MPI-PKS, 2015; QUTIF Research School, Rostock, 2016
- Spin transfer in bicircular HHG: a photon exchange model. CORINF 2014 Summer School, Cargèse, 2014; GRC Multiphoton Processes, Waltham, 2014; Quantum Optics VII, Mar del Plata, 2014
- Angular distributions for correlation-assisted tunnelling. ICOLS 2013, Berkeley; GRC Quantum Control of Light and Matter, Mount Holyoke College; ATTO2013, Paris, 2013
- Channel jumping inside a tunnel ionization barrier. ATTOFEL Winter School, Bormio, 2013

- Under-the-barrier electron-ion interaction during tunnel ionization. QuICC 2012, Aberystwyth, 2012
- Quadrature (pseudo)eigenstates for finite photon numbers. QuICC 2012. Aberystwyth, 2012
- Analytical Structure of the Annihilation Operator. IV Annual Meeting of the Quantum Information Division (DICU), Querétaro, 2011.